Question ID: ee41d7e0

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Easy

Question

Arthropods—brine shrimp, hawk moths, and many other invertebrate animals—have a nervous system made up of a brain, nerve cord, and other nerves. Researchers have gained insights about this system in ancient arthropods from traces found in various fossils. For example, in a study of two fossils of the extinct arthropod species *Mollisonia symmetrica*, Javier Ortega-Hernández, James Weaver, and team observed clear signs of a nerve cord. They also saw possible indications of a synganglion, a brain-like mass of nerves. Researchers hope to identify more features of the nervous systems of prehistoric arthropods as additional fossils are found.

Which choice best states the main idea of the text?

Answer

A. There are several similarities between the brains of hawk moths and the brains of brine shrimp.

B. Fossil evidence can contribute to the understanding of the nervous system in ancient arthropods.

C. Newly discovered fossils suggest that ancient hawk moths and ancient brine shrimp had spines.

D. Researchers need to focus on finding more fossils of ancient arthropods.

Correct Answer: B

Rationale

Choice B is the best answer because it most accurately states the main idea of the text. The text states that researchers have used fossils to learn about the nervous systems of ancient arthropods. It then provides a specific example of researchers studying fossils of *Mollisonia symmetrica* and finding evidence of a nerve cord and possibly a synganglion (a brain-like mass of nerves). The text concludes by noting that researchers hope to discover more features of prehistoric arthropods' nervous systems "as additional fossils are found." Thus, the main idea of the text is that fossil evidence can contribute to the understanding of the nervous system in ancient arthropods.

Choice A is incorrect because the text doesn't compare the brains of hawk moths and brine shrimp. These animals are merely mentioned as examples of arthropods in the opening sentence, and the text doesn't go on to discuss any similarities between their brains. Choice C is incorrect because the text discusses nervous systems, not spines, in ancient arthropods, and it doesn't specifically mention findings about ancient hawk moths or brine shrimp. Choice D is incorrect. While the text concludes by noting that researchers aim to discover additional characteristics of prehistoric arthropods' nervous systems as more fossils are uncovered, this statement doesn't suggest that researchers need to focus on finding more fossils. Rather, it simply indicates that researchers expect to learn more about this topic as additional fossils are discovered. Moreover, the statement is just a detail that supports the main idea; it's not the main idea itself.